AROR UNIVERSITY OF ART, ARCHITECTURE, DESIGN & HERITAGE, SUKKUR, SINDH

Course: Applied Physics

Lecture 01

Instructor:

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Course Outlines

1. Mechanics

- Units, Physical Quantities, and Vectors
- Motion Along a Straight Line
- Motion in Two or Three Dimensions
- Newton's Laws of Motion
- Applying Newton's Laws
- Work and Kinetic Energy
- Potential Energy and Energy Conservation
- Momentum, Impulse, and Collisions
- Rotation of Rigid Bodies
- Dynamics of Rotational Motion
- Equilibrium and Elasticity
- Fluid Mechanics
- Gravitation
- Periodic Motion

2. Waves/Acoustics

- Mechanical Waves
- Sound and Hearing

3. Electromagnetism

- Electric Charge and Electric Field
- Gauss's Law
- Electric Potential
- Capacitance and Dielectrics
- Current, Resistance, and Electromotive Force
- Direct-Current Circuits
- Magnetic Field and Magnetic Forces
- Sources of Magnetic Field
- Electromagnetic Induction
- Inductance
- Alternating Current
- Electromagnetic Waves

Recommended Books

S.No	Book Name	Author/s Name	Publisher Name & Edition
1.	Fundamentals of Physics	David Halliday, Robert Resnick, and Jearl Walker	Ninth Edition, John Wiley & Sons, ISBN: 0471465097.
2.	University Physics	Hugh D Young and Roger A. Freedman	14th Edition
3.	College Physics	Asif, R. (2017).	Intermediate-1, Pearl Educational, publishing’s, urdu Bazar. Lahore, Pakistan.

Assessment Criteria

1.	Sessional (Quiz, Assignments, Presentations & Class response)	30%
2.	Mid term	30%
3.	Final Examination	40%

Start of Lecture 01

Contents

1
Physics

Definition:

“Physics is the branch of science that studies matter, energy, space, time, and the interactions between them, in order to understand how the universe behaves.”

- Physics is one of the most fundamental sciences.
- Ideas of physics are used by scientists in many fields such as chemistry, paleontology, and climatology.
- Physics forms the foundation of all engineering and technology.
- Engineering designs (TVs, spacecraft, tools, etc.) require understanding basic physical laws.
- Studying physics is challenging but also rewarding and intellectually satisfying.
- Physics explains everyday and natural phenomena (blue sky, radio waves, satellite motion).
- Physics represents a major achievement of human intellect in understanding nature.

Mechanics

Mechanics

- Mechanics is a branch of physics that deals with the motion of objects and the forces that cause or change that motion.
- Mechanics explains how and why objects move or stay at rest.

What Does Mechanics Study?

Mechanics mainly answers three questions:

1. **Where is the object?** (position)
2. **How is it moving?** (motion)
3. **Why is it moving or not moving?** (force)

Main Branches of Mechanics

1. Statics

- Studies objects **at rest**
- Example: A book lying on a table

2. Kinematics

- Studies **motion without considering forces**
- Example: Speed of a moving car

3. Dynamics

- Studies **motion along with forces**
- Example: Why a ball speeds up when kicked

Mechanics (2)

Examples from Daily Life

- ✔ A car moving on the road
- ✔ A fan rotating
- ✔ A stone falling from a height
- ✔ Pushing a box on the floor

All these are studied in **mechanics**.

Why Is Mechanics Important?

- ✔ Forms the **foundation of physics**
- ✔ Used in engineering, machines, and structures
- ✔ Helps understand everyday motions

One-Line Definition

📌 *Mechanics is the branch of physics that studies motion, rest, and the forces acting on bodies.*

3
Standards and units

Fundamental Quantities	SI Unit	Symbol
Length	meter	m
Mass	kilogram	kg
Time	second	S
Electric current	Ampere	A
Temperature	Kelvin	K
Amount of subtance	mol	mol
Luminous Intensity	candela	cd

4
Unit Prefixes

Submult	Prefix	Char	Mult	Prefix	Char
10^{-1}	deci	d	10	deca	da
10^{-2}	centi	c	10^2	hecto	h
10^{-3}	milli	m	10^3	kilo	k
10^{-6}	micro	u	10^6	mega	M
10^{-9}	nano	n	10^9	giga	G
10^{-12}	pico	p	10^{12}	tera	T
10^{-15}	femto	f	10^{15}	peta	P
10^{-18}	atto	a	10^{18}	exa	E
10^{-21}	zepto	z	10^{21}	zetta	Z
10^{-24}	yocto	y	10^{24}	yotta	Y

1 kilometer = 1 km = 10^3 meters = 10^3 m

1 kilogram = 1 kg = 10^3 grams = 10^3 g

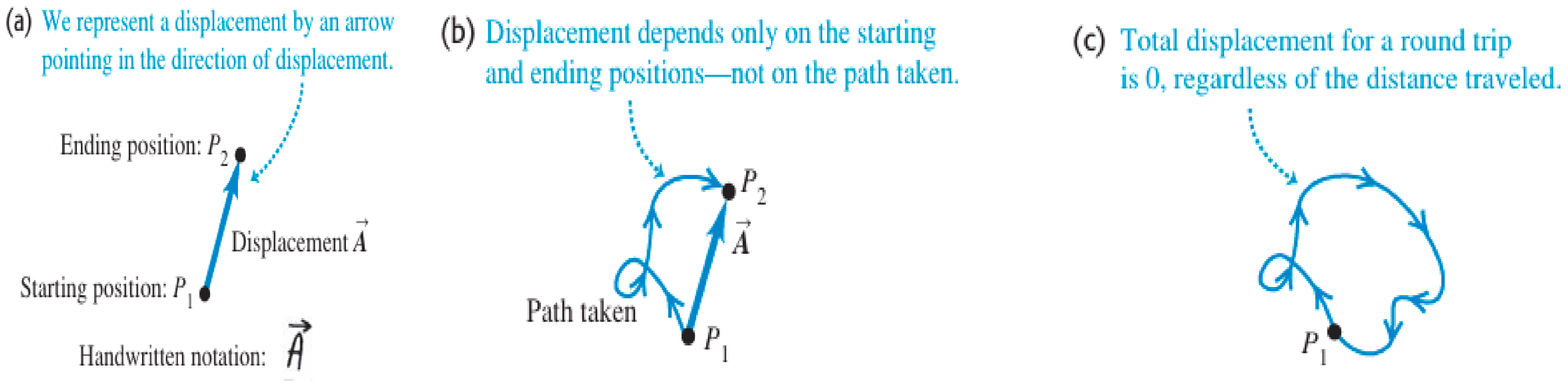
1 kilowatt = 1 kW = 10^3 watts = 10^3 W

5
Scalars and Vectors

- Some physical quantities, such as time, temperature, mass, and density, can be described completely by a single number with a unit.
- But many other important quantities in physics have a *direction* associated with them and cannot be described by a single number.
  - A simple example is the motion of an airplane: We must say not only how fast the plane is moving but also in what direction. The speed of the airplane combined with its direction of motion constitute a quantity called *velocity*.
  - Another example is *force,* which in physics means a push or pull exerted on an object. Giving a complete description of a force means describing both how hard the force pushes or pulls on the object and the direction of the push or pull.
- When a physical quantity is described by a *single number,* we call it a **scalar quantity.**
- In contrast, a **vector quantity** has both a **magnitude** (the "how much" or "how big" part) and a direction in space.
- Calculations that combine scalar quantities use the operations of ordinary arithmetic.
  - For example, 6 kg + 3 kg = 9 kg, or 4 * 2 s = 8 s.
- However, combining vectors requires a different set of operations.

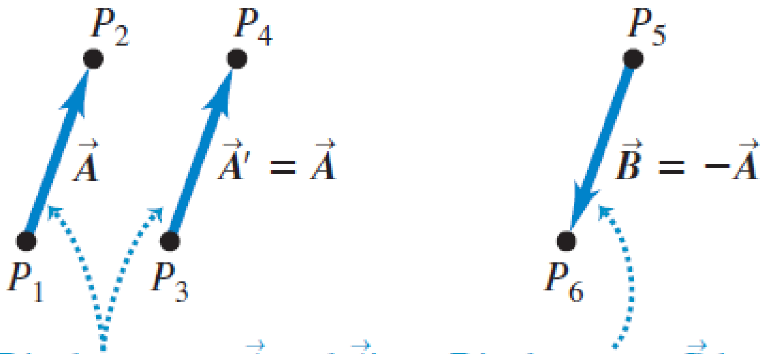
Scalars and Vectors (3)

- Understanding the Vector:
  - Displacement as a vector quantity.
  - A displacement is always a straight-line segment directed from the starting point to the ending point, even if the path is curved.



Scalars and Vectors (4)

- The meaning of vectors that have the same magnitude and the same or opposite direction:

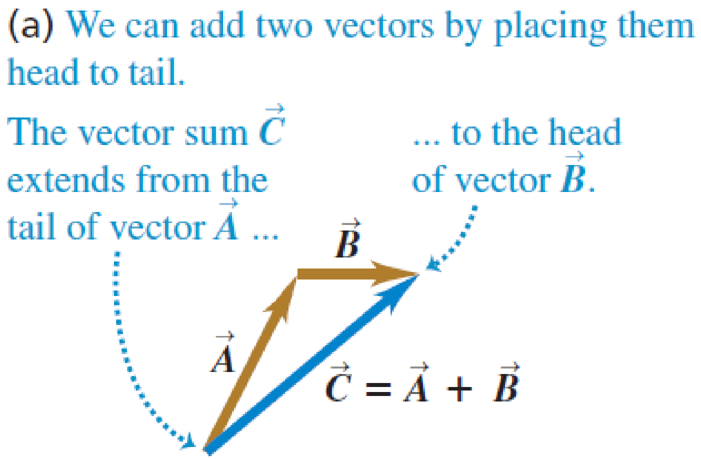


Displacements $\vec{A}$ and $\vec{A}'$ are equal because they have the same length and direction.

Displacement $\vec{B}$ has the same magnitude as $\vec{A}$ but opposite direction; $\vec{B}$ is the negative of $\vec{A}$.

6
Vector Addition and Subtraction

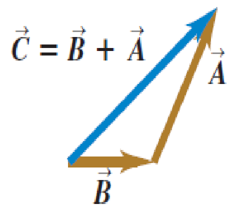
- To understand more about vectors and how they combine, we start with the simplest vector quantity, **displacement.**
- Suppose a particle undergoes a displacement ***A*** followed by a second displacement ***B***.
- The final result is the same as if the particle had started at the same initial point and undergone a single displacement ***C*** as shown in fig.
- We call displacement ***C*** the **vector sum,** or **resultant,** of displacements ***A*** and ***B.***
- In vector addition we usually place the *tail* of the *second* vector at the *head*, or tip, of the *first* vector,



Vector Addition and Subtraction (2)

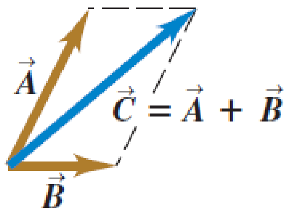
- If we make the displacements ***A*** and ***B*** in reverse order, with ***B*** first and ***A*** second, the result is the same.
- Thus, $\vec{C} = \vec{B} + \vec{A}$ and $\vec{A} + \vec{B} = \vec{B} + \vec{A}$
- This shows that the order of terms in a vector sum doesn't matter.
- In other words, vector addition obeys the *commutative* law.

(b) Adding them in reverse order gives the same result: $\vec{A} + \vec{B} = \vec{B} + \vec{A}$. The order doesn't matter in vector addition.



- *Figure (c)* shows another way to represent the vector sum:
- If we draw vectors ***A*** and ***B*** with their tails at the same point, vector ***C*** is the diagonal of a parallelogram constructed with ***A*** and ***B*** as two adjacent sides.

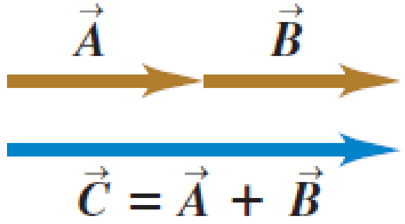
(c) We can also add two vectors by placing them tail to tail and constructing a parallelogram.



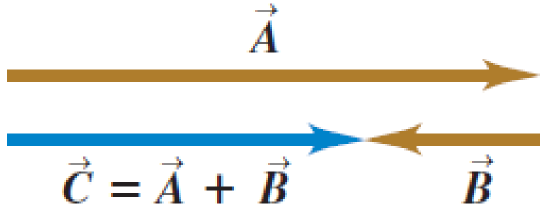
Vector Addition and Subtraction (3)

Adding vectors that are (a) parallel and (b) antiparallel:

(a) Only when vectors $\vec{A}$ and $\vec{B}$ are parallel does the magnitude of their vector sum $\vec{C}$ equal the sum of their magnitudes: $C = A + B$.



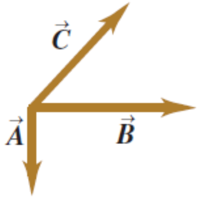
(b) When $\vec{A}$ and $\vec{B}$ are antiparallel, the magnitude of their vector sum $\vec{C}$ equals the *difference* of their magnitudes: $C = |A - B|$.



Vector Addition and Subtraction (4)

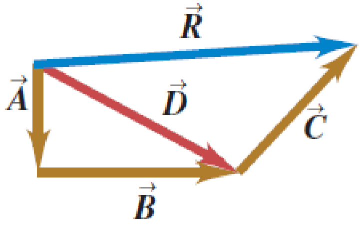
- **Figure a** shows *three* vectors ***A*** , ***B*** , and ***C.***

(a) To find the sum of these three vectors ...

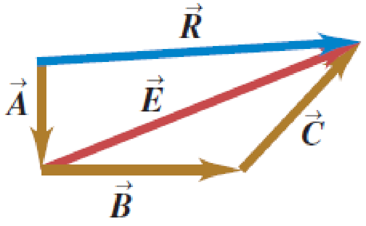


Several constructions for finding the vector sum $\vec{A} + \vec{B} + \vec{C}$.

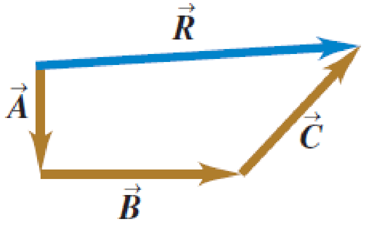
(b) ... add $\vec{A}$ and $\vec{B}$ to get $\vec{D}$ and then add $\vec{C}$ to $\vec{D}$ to get the final sum (resultant) $\vec{R}$ ...



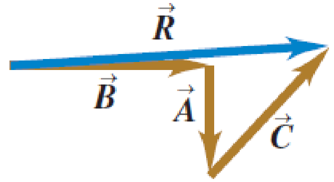
(c) ... or add $\vec{B}$ and $\vec{C}$ to get $\vec{E}$ and then add $\vec{E}$ to $\vec{A}$ to get $\vec{R}$ ...



(d) ... or add $\vec{A}$, $\vec{B}$, and $\vec{C}$ to get $\vec{R}$ directly ...



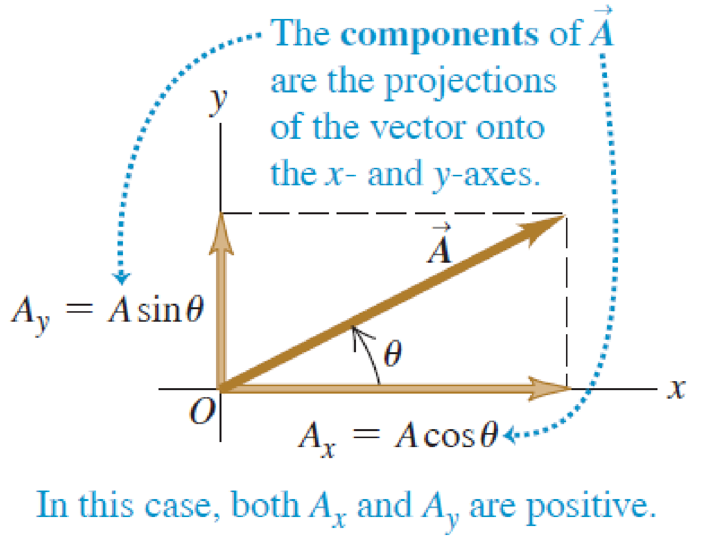
(e) ... or add $\vec{A}$, $\vec{B}$, and $\vec{C}$ in any other order and still get $\vec{R}$.



Components of Vectors

- Previously, we added vectors by using a scale diagram and properties of right triangles.
- But calculations with right triangles work only when the two vectors are perpendicular.
- So we need a simple but general method for adding vectors. This is called the *method of components.*
- To define what we mean by the components of a vector $\boldsymbol{A}$, we begin with a rectangular (Cartesian) coordinate system of axes (**as shown in Fig**). If we think of $\boldsymbol{A}$ as a displacement vector, we can regard $\boldsymbol{A}$ as the sum of a displacement parallel to the *x*-axis and a displacement parallel to the *y*-axis.
- We use the numbers *Ax* and *Ay* to tell us how much displacement there is parallel to the *x*-axis and how much there is parallel to the *y*-axis, respectively.

Figure 1.17 Representing a vector $\vec{A}$ in terms of its components A_x and A_y.



Components of Vectors (2)

For example:

- If the +x-axis points east and the +y-axis points north, $\boldsymbol{A}$ (in Fig at previous slide) could be the sum of a 2.00 m displacement to the east and a 1.00 m displacement to the north.
- Then Ax = +2.00 m and Ay = +1.00 m.
- We can use the same idea for any vectors, not just displacement vectors.
- The two numbers Ax and Ay are called the **components** of vector $\boldsymbol{A}$.

CAUTION **Components are not vectors** The components A_x and A_y of a vector $\vec{A}$ are numbers; they are *not* vectors themselves. This is why we print the symbols for components in lightface italic type with *no* arrow on top instead of in boldface italic with an arrow, which is reserved for vectors.

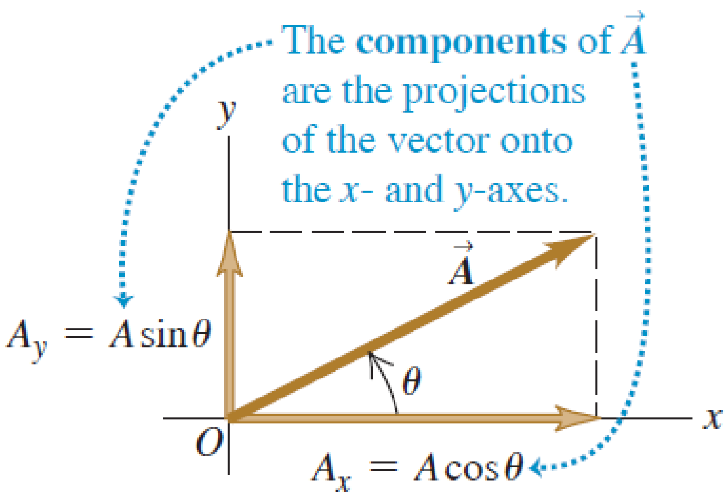
Components of Vectors (3)

- We can calculate the components of vector $\overrightarrow{A}$ if we know its magnitude A and its direction.
- We'll describe the direction of a vector by its angle relative to some reference direction.
- In Fig. this reference direction is the positive x-axis, and the angle between vector $\vec{A}$ and the positive x-axis is θ (the Greek letter theta).
-

$$\frac{A_x}{A} = \cos\theta \qquad \text{and} \qquad \frac{A_y}{A} = \sin\theta$$

$$A_x = A\cos\theta \qquad \text{and} \qquad A_y = A\sin\theta$$

(θ measured from the $+x$-axis, rotating toward the $+y$-axis)



In this case, both A_x and A_y are positive.

Components of Vectors (4)

Using Components to Do Vector Calculations

- Using components makes it relatively easy to do various calculations involving vectors.
- Let's look at three important examples:
  - Finding a vector's magnitude and direction,
  - Multiplying a vector by a scalar, and
  - Calculating the vector sum of two or more vectors.

1. Finding a vector's magnitude and direction from its components:

By applying the Pythagorean theorem to Fig., we find that the magnitude of vector $\vec{A}$ is

$$A = \sqrt{A_x^2 + A_y^2}$$

$$\tan\theta = \frac{A_y}{A_x} \quad \text{and} \quad \theta = \arctan\frac{A_y}{A_x}$$

Components of Vectors (5)

II. Multiplying a vector by a scalar:

➢ If we multiply a vector $\vec{A}$ by a scalar c, each component of the product $\vec{D} = c\,\vec{A}$ is the product of c and the corresponding component of $\vec{A}$:

$$D_x = cA_x, \qquad D_y = cA_y \quad (\text{components of } \vec{D} = c\vec{A})$$

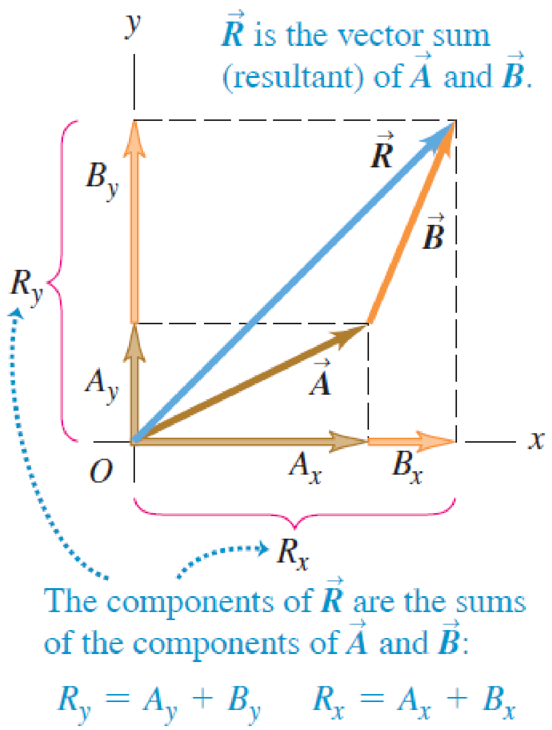
III. Using components to calculate the vector sum (resultant) of two or more vectors:

Each component of $\vec{R} = \vec{A} + \vec{B}$ ...

$$R_x = A_x + B_x, \qquad R_y = A_y + B_y$$

... is the sum of the corresponding components of $\vec{A}$ and $\vec{B}$.

Figure 1.21 Finding the vector sum (resultant) of $\vec{A}$ and $\vec{B}$ using components.



Numerical-I

Problem (Practical Example)

A **force** acting on an object has the following components:

- Horizontal component $F_x = 6\,\mathrm{N}$ (towards the east)
- Vertical component $F_y = 8\,\mathrm{N}$ (upward)

Find:

1. The **magnitude** of the force
2. The **direction** of the force with respect to the east direction

Numerical-I (cont.)

Solution

Step 1: Find the Magnitude of the Force

The magnitude of a vector is:

$$F = \sqrt{F_x^2 + F_y^2}$$

$$F = \sqrt{6^2 + 8^2}$$

$$F = \sqrt{36 + 64}$$

$$F = \sqrt{100}$$

$$F = 10\,\mathrm{N}$$

👉 **The total force acting on the object is 10 N.**

Step 2: Find the Direction of the Force

$$\theta = \tan^{-1}\left(\frac{F_y}{F_x}\right)$$

$$\theta = \tan^{-1}\left(\frac{8}{6}\right)$$

$$\theta \approx 53.1^\circ$$

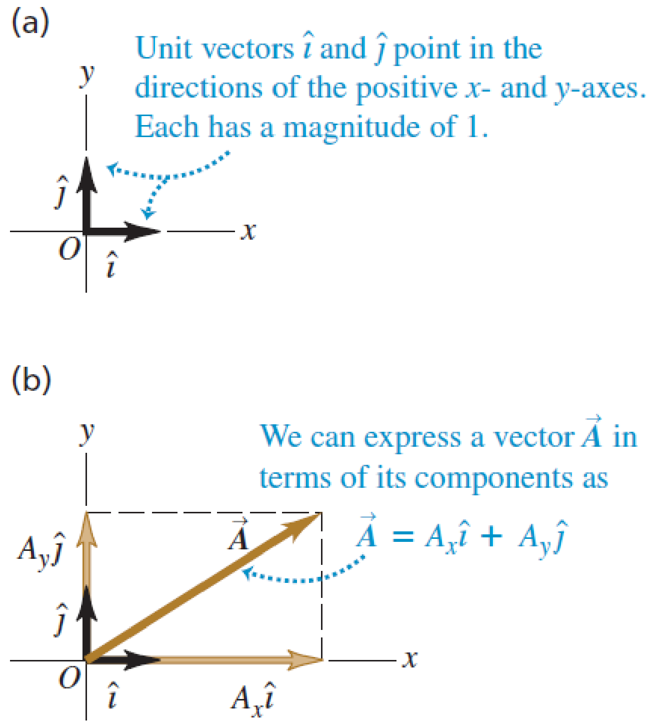
👉 **The force acts at an angle of 53.1° above the east direction.**

8
Unit Vectors

- A **unit vector** is a vector that has a magnitude of 1, with no units.
- Its only purpose is to *point*—that is, to describe a direction in space.
- Unit vectors provide a convenient notation for many expressions involving components of vectors.
- We'll always include a caret, or "hat" (^), in the symbol for a unit vector to distinguish it from ordinary vectors whose magnitude may or may not be equal to 1.
- In an *xy*-coordinate system we can define a unit vector $\hat{\imath}$ that points in the direction of the positive *x*-axis and a unit vector $\hat{\jmath}$ that points in the direction of the positive *y*-axis (**Fig. 1.24a**).

$$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath}$$

Figure **1.24** (a) The unit vectors $\hat{\imath}$ and $\hat{\jmath}$. (b) Expressing a vector $\vec{A}$ in terms of its components.



Unit Vectors (2)

Using unit vectors, we can express the vector sum $\vec{R}$ of two vectors $\vec{A}$ and $\vec{B}$ as follows:

$$
\begin{aligned}
\vec{A} &= A_x\hat{\imath} + A_y\hat{\jmath} \\
\vec{B} &= B_x\hat{\imath} + B_y\hat{\jmath} \\
\vec{R} &= \vec{A} + \vec{B} \\
&= (A_x\hat{\imath} + A_y\hat{\jmath}) + (B_x\hat{\imath} + B_y\hat{\jmath}) \\
&= (A_x + B_x)\hat{\imath} + (A_y + B_y)\hat{\jmath} \\
&= R_x\hat{\imath} + R_y\hat{\jmath}
\end{aligned}
$$

Unit vectors (3)

- If not all of the vectors lie in the *xy*-plane, then we need a third component.
- We introduce a third unit vector $\hat{k}$ that points in the direction of the positive *z*-axis (**Fig. 1.25**).
-

Any vector can be expressed in terms of its *x*-, *y*-, and *z*-components ...

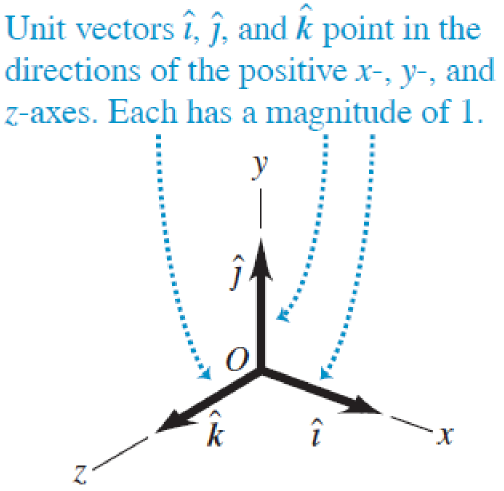
$$\vec{A} = A_x\hat{\imath} + A_y\hat{\jmath} + A_z\hat{k}$$

$$\vec{B} = B_x\hat{\imath} + B_y\hat{\jmath} + B_z\hat{k}$$

... and unit vectors $\hat{\imath}$, $\hat{\jmath}$, and $\hat{k}$.

$$\vec{R} = (A_x + B_x)\hat{\imath} + (A_y + B_y)\hat{\jmath} + (A_z + B_z)\hat{k}$$
$$= R_x\hat{\imath} + R_y\hat{\jmath} + R_z\hat{k}$$

Figure 1.25 The unit vectors $\hat{\imath}$, $\hat{\jmath}$, and $\hat{k}$.



Numerical-II

Given the two displacements

$$\vec{D} = (6.00\hat{\imath} + 3.00\hat{\jmath} - 1.00\hat{k})\ \text{m} \quad \text{and}$$

$$\vec{E} = (4.00\hat{\imath} - 5.00\hat{\jmath} + 8.00\hat{k})\ \text{m}$$

find the magnitude of the displacement $2\vec{D} - \vec{E}$.

$$\vec{F} = 2(6.00\hat{\imath} + 3.00\hat{\jmath} - 1.00\hat{k})\ \text{m} - (4.00\hat{\imath} - 5.00\hat{\jmath} + 8.00\hat{k})\ \text{m}$$

$$= [(12.00 - 4.00)\hat{\imath} + (6.00 + 5.00)\hat{\jmath} + (-2.00 - 8.00)\hat{k}]\ \text{m}$$

$$= (8.00\hat{\imath} + 11.00\hat{\jmath} - 10.00\hat{k})\ \text{m}$$

From Eq. (1.12) the magnitude of $\vec{F}$ is

$$F = \sqrt{F_x{}^2 + F_y{}^2 + F_z{}^2}$$

$$= \sqrt{(8.00\ \text{m})^2 + (11.00\ \text{m})^2 + (-10.00\ \text{m})^2}$$

$$= 16.9\ \text{m}$$

Numerical-III

Arrange the following vectors in order of their magnitude, with the vector of largest magnitude first.

(i) A = (3i+5j - 2k) m

(ii) B = (-3i + 5j - 2k)m

(iii) C = (3i - 5j -2k) m

(iv) D = (3i+ 5J + 2k) m.

To arrange the vectors by magnitude, we calculate the magnitude of each vector using:

$$|\vec{V}| = \sqrt{x^2 + y^2 + z^2}$$

$$\vec{A} = (3i + 5j - 2k)$$

$$|\vec{A}| = \sqrt{3^2 + 5^2 + (-2)^2} = \sqrt{9 + 25 + 4} = \sqrt{38}$$

$$\vec{B} = (-3i + 5j - 2k)$$

$$|\vec{B}| = \sqrt{(-3)^2 + 5^2 + (-2)^2} = \sqrt{38}$$

$$\vec{C} = (3i - 5j - 2k)$$

$$|\vec{C}| = \sqrt{3^2 + (-5)^2 + (-2)^2} = \sqrt{38}$$

$$\vec{D} = (3i + 5j + 2k)$$

$$|\vec{D}| = \sqrt{3^2 + 5^2 + 2^2} = \sqrt{38}$$

$$\boxed{A = B = C = D}$$

All four vectors have **equal magnitude**.